

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

*Bilimingga ishon va muvaffaqiyatga erish!*

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY  
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA  
FANIDAN  
TEST TOPSHIRIQLARI KITOBIDIR

*Ishtirokchining familiyasi, ismi va otasining ismi*

*Imzo*

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath\_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'rinishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

## MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!

1.03.2026. 2-smena.

1. Olti xonali  $\overline{8629ab}$  soni 12 ga qoldiqsiz bo'lsa,  $a \cdot b$  eng katta qiymati toping.

- A) 48
- B) 56
- C) 72
- D) 54

2. Hisoblang (Bu yerda,  $[a]$  -  $a$  sonning butun qismi,  $\{a\}$  -  $a$  sonining kasr qismi.):

$$\left\{ 3[\lg 5] - 5 \left\{ \lg \frac{1}{10} \right\} + 7[\ln 3] \right\}.$$

- A) 0,2
- B) 0,1
- C) 0,7
- D) 0

3.  $A$  shahardan  $B$  shaharga qarab piyoda yo'lga chiqdi. Undan 3 soat o'tib velosipedchi yo'lga chiqdi. Velosipedchi tezligi piyodaning tezligidan 4 barobar katta. Velosipedchi piyodaga yo'lning o'rtasida yetib oldi. Piyoda  $A$  shahardan  $B$  shaharga necha soatda yetib boradi?

- A) 10
- B) 8
- C) 12
- D) 6

4. Sotuvchi shim va ko'ylak sotmoqda. Ko'ylak 45%, shim 40% oshirilib bittadan sotilganda 215000 so'm foyda, shim 45% ko'ylak 40% oshirilib sotilganda 210000 so'm foyda qolasa, ko'ylakning narxini toping.

- A) 180000
- B) 200000
- C) 300000
- D) 250000

5. Hisoblang:

$$(1 + (2 - (3 + (4 - 1)^{-1})^{-1})^{-1})^{-1}.$$

- A)  $\frac{17}{27}$
- B)  $\frac{17}{10}$
- C)  $\frac{27}{17}$
- D)  $\frac{10}{17}$

6. Hisoblang:

$$\left(9,4\sqrt[3]{6^2 \cdot \sqrt{6}} - 3,4\sqrt{6 \cdot \sqrt[3]{6^2}}\right)^{\frac{6}{11}}.$$

- A) 3
- B) 6
- C) 5
- D) 9

7. Hisoblang:

$$\sqrt[3]{26 + 15\sqrt{3}} + \sqrt[3]{26 - 15\sqrt{3}}.$$

- A) 1
- B) 2
- C) 3
- D) 4

8.  $a_n$  arifmetik progressiyada  $a_2 = 2a_1$  va  $a_{10} = 20$  bo'lsa,  $S_6 - S_5$  ni toping.

- A) 16
- B) 10
- C) 12
- D) 14

9.  $b_1 + b_2 + b_3 = 217$  va  $b_1 = a_2$ ,  $b_2 = a_9$ ,  $b_3 = a_{44}$  hamda  $a_1 + a_2 + \dots + a_n = 820$  bo'lsa  $n$  ni toping. Bu yerda  $a_n$  arifmetik progressiya,  $b_n$  geometrik progressiya.

- A) 20
- B) 19
- C) 18
- D) 21

10.  $y \neq -1$  uchun  $\frac{x-2}{y} = 1$  bo'lsa,  $\frac{2x+4y+2}{x+y}$  ning qiymatini toping.

- A) 1
- B)  $\frac{1}{3}$
- C) 3
- D) 2

11. Soddalashtiring:

$$\frac{1}{a + \frac{1}{b + \frac{1}{c}}} : \frac{1}{a + \frac{1}{b}} - \frac{1}{b \cdot (abc + a + c)}$$

- A)  $a$
- B)  $b$

C)  $ab$ 

D) 1

12. Hisoblang:

$$\frac{\sin 11^\circ - \sin 49^\circ}{1 - 2 \cos^2 54,5} .$$

A)  $\sqrt{3}$ B)  $\frac{1}{\sqrt{3}}$ C)  $-\sqrt{3}$ D)  $-\frac{1}{\sqrt{3}}$ 

13. Tengsizlikni yeching:

$$(0, 75)^{\frac{x^2+x-8}{x+1}} > 0,6 \cdot (1, 25)^{x-\frac{8}{x+1}} .$$

A)  $(-\infty; -3) \cup (-1; 3)$ B)  $(-\infty; -3) \cup (-3; 3)$ C)  $(-\infty; -3) \cup (-3; 1)$ D)  $(-\infty; -3) \cup (-3; -1)$ 

14. Tengsizlikni nechta butun yechimi bor:

$$\log_{0,5} \log_2 \log_{x-1} 9 > 0 .$$

A) 4 ta

B) 5 ta

C) 6 ta

D) 7 ta

15. Tenglamaning  $\left[-\frac{5\pi}{2}; 2\pi\right]$  oralig'idagi yechimlari sonini toping:

$$\sqrt{3} \sin^2 x + \sin^2 2x = \frac{\sqrt{3}}{2} - \frac{1}{2} - \sqrt{3} \cos^2 x .$$

A) 1 ta

B) 3 ta

C) Haqiqiy yechimga ega emas

D) 4 ta

16. Tenglama nechta haqiqiy ildizga ega:

$$8x^3 - 6x^2 + 3x - 1 = 0 .$$

A) 1 ta

B) 2 ta

C) 3 ta

D) Haqiqiy yechimga ega emas

17. Tengsizlikni yeching:

$$\frac{1}{x+1} + \frac{2}{x+2} - \frac{3}{x+3} > 0.$$

- A)  $(-\infty; -3) \cup (-2; -1, 5) \cup (1; \infty)$
- B)  $(-\infty; -3) \cup (-2; -1) \cup (-1; \infty)$
- C)  $(-\infty; -3) \cup (-2; -1, 5) \cup (-1; \infty)$
- D)  $(-\infty; -3) \cup (-2; 1) \cup (1; \infty)$

18. Tenglamaning haqiqiy ildizlari soni nechta:

$$|x^2 - 5x - 6| + |7 + 6x - x^2| = x + 1.$$

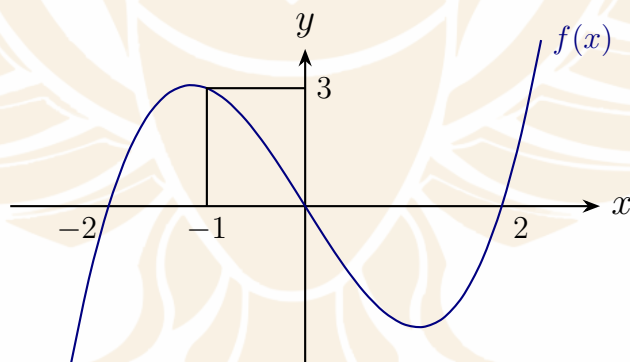
- A) 3 ta
- B) Cheksiz ko'p yechimga ega
- C) Haqiqiy ildizga ega emas
- D) 5 ta

19. Tengsizlikning butun yechimlar sonini toping:

$$(2x^2 - x - 2)^2 < (x^2 + 5x + 4)^2.$$

- A) 7 ta
- B) 6 ta
- C) 8 ta
- D) 5 ta

20.  $f(x) = ax^3 + bx^2 + cx + d$  funksiyaning grafigi tasvirlangan.  $f(-1) = 3$  bo'lsa,  $f(-3)$  ni toping.



- A) 12
- B) -12
- C) -15
- D) 15

21.  $y = f(x)$  funksiya juft funksiya va  $y = g(x)$  funksiya esa toq funksiya bo'lsin. Quyidagilardan nechitasi toq funksiya bo'ladi:

1.  $y = f(x) \cdot g(x);$
2.  $y = \frac{f(x)}{g(x)};$

3.  $y = f(x) + g(x)$ .

- A) 0 ta
- B) 1 ta
- C) 2 ta
- D) 3 ta

22.  $f(x) = |\ln x - \log_{\cos x} \sin x|$  ning  $x_0 = \frac{\pi}{4}$  nuqtadagi hosilasini toping.

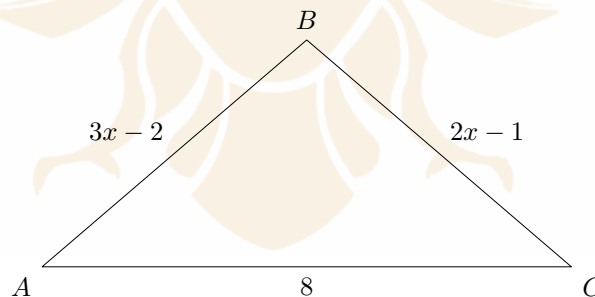
- A)  $-\frac{4}{\pi} - \frac{4}{\ln 2}$
- B)  $\frac{4}{\pi} - \frac{4}{\ln 2}$
- C)  $-\frac{4}{\pi} + \frac{4}{\ln 2}$
- D)  $\frac{4}{\pi} + \frac{4}{\ln 2}$

23. Aniqmas integralni toping:

$$\int \frac{x}{\sqrt{3x+1}} dx .$$

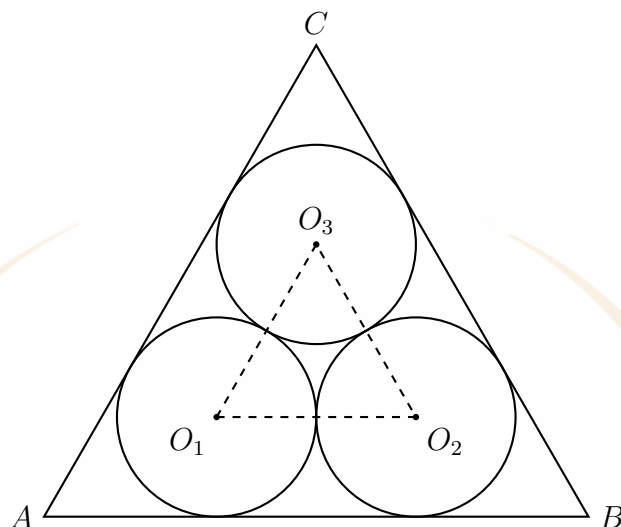
- A)  $\frac{2}{9} \left( \frac{1}{3}(3x+1)^{\frac{3}{2}} + \sqrt{3x+1} \right) + C$
- B)  $\frac{2}{9} \left( \frac{1}{3}(3x+1)^{\frac{3}{2}} - \sqrt{3x+1} \right) + C$
- C)  $\frac{2}{3} \left( \frac{1}{3}(3x+1)^{\frac{3}{2}} - \sqrt{3x+1} \right) + C$
- D)  $\frac{2}{3} \left( \frac{1}{3}(3x+1)^{\frac{3}{2}} + \sqrt{3x+1} \right) + C$

24. Tomonlari har xil butun sonlardan iborat uchburchakning eng katta tomoni(asosi) 8 ga teng. Yon tomonlari esa  $(3x-2)$  va  $(2x-1)$  ga teng. Perimetrini eng katta qiymatini toping.

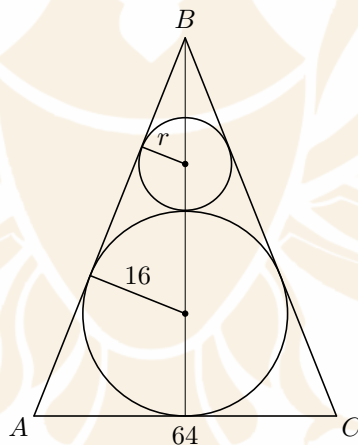


- A) 18
- B) 20
- C) 21
- D) 22

25.  $ABC$  muntazam uchburchak ichiga teng radiusli bir biriga urinuvchi 3 ta aylana chizilgan. Shu aylana markazlari tutashishidan  $O_1O_2O_3$  uchburchak hosil qilingan.  $ABC$  uchburchak yuzini  $O_1O_2O_3$  uchburchak yuziga nisbatini toping.

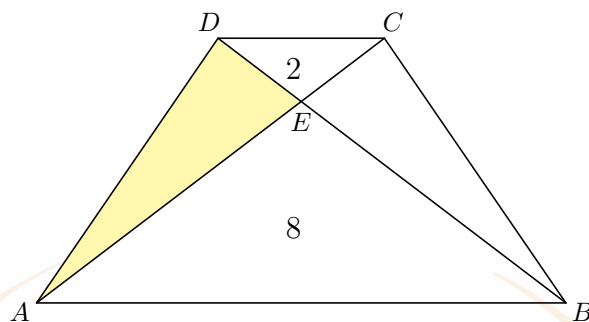


- A)  $2 + \sqrt{3}$   
 B)  $2 + 2\sqrt{3}$   
 C)  $4 + 2\sqrt{3}$   
 D) 4
26.  $ABC$  teng yonli uchburchakda  $AB = BC$  va  $AC = 64$  ga teng. Uchburchakka radiuslari  $r$  va 16 ga teng ayanalar ichki chizilgan.  $r$  ni toping.



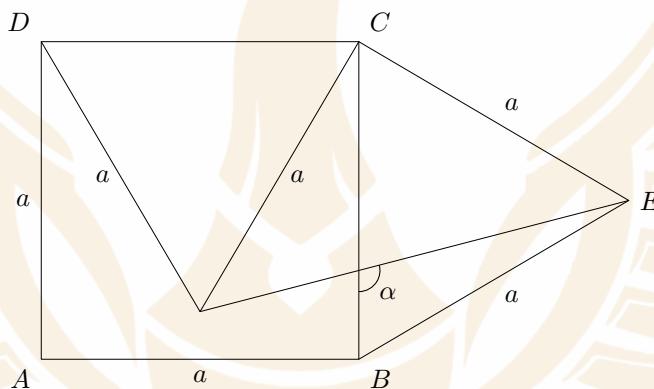
- A) 6  
 B) 2  
 C) 4  
 D) 12

27.  $ABCD$  trapetsiyaning dioganallari o'tkazilgan.  $S_{DEC} = 2$  va  $S_{AEB} = 8$  bo'lsa,  $S_{AED}$  ni toping.



- A) 6
- B) 4
- C) 2
- D) 12

28. Quyidagi chizmada  $ABCD$  kvadrat va  $CBE$  muntazam uchburchak tasvirlangan. Shunga ko'ra  $\alpha$  burchakni toping.



- A)  $150^\circ$
- B)  $100^\circ$
- C)  $105^\circ$
- D)  $90^\circ$

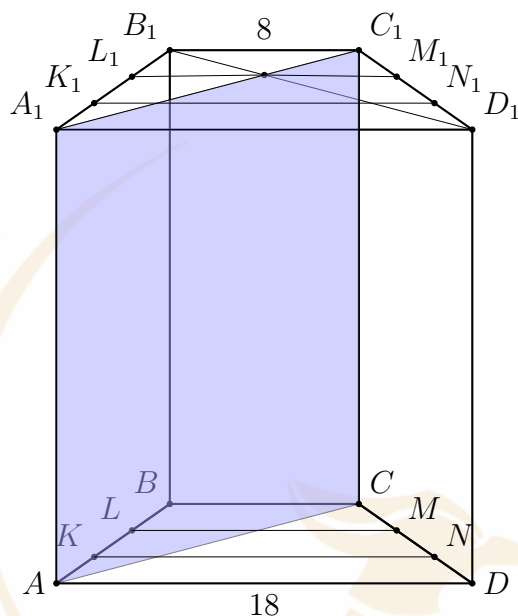
29. Uchlari  $A(1, 4, 2)$ ,  $B(3, 2, 2)$  va  $C(7, 2, 5)$  da bo'lgan uchburchak berilgan.  $BD$  medianasi va  $CB$  tomon orasidagi burchakni sinusini toping.

- A)  $\frac{\sqrt{7}}{5}$
- B)  $\frac{2\sqrt{2}}{5}$
- C)  $\frac{3\sqrt{2}}{5}$
- D)  $\frac{\sqrt{2}}{5}$

30. Teng yonli to'g'ri burchakli uchburchakning  $BC$  kateti  $\alpha$  tekislikda yotadi.  $AC$  kateti  $\alpha$  tekislik bilan  $45^\circ$  hosil qiladi.  $\alpha$  tekislik va gipotenuza orasidagi burchakni toping.
- A)  $45^\circ$   
B)  $60^\circ$   
C)  $30^\circ$   
D)  $\arcsin \frac{1}{\sqrt[4]{2}}$
31. 25 nafar o'quvchi bor. 9 ta o'quvchi biologiya to'garagiga boradi. 8 ta o'quvchi kimyo to'garagiga boradi. 10 ta o'quvchi hech qaysisiga bormaydi. Nechta o'quvchi 2 ta to'garakkayam boradi.
- A) 1  
B) 2  
C) 3  
D) 4
32.  $A$  va  $B$  to'g'ri chiziqlar kesishmaydi.  $A$  dan 3 ta nuqta,  $B$  da 4 nuqta olingan. Bu nuqtalar orqali nechta uchburchak yasash mumkin?
- A) 24 ta  
B) 48 ta  
C) 30 ta  
D) 36 ta

Topshiriqlar (33–35) va javob variant (A–F) larni o‘zaro moslashtiring.

Prizma asosi teng yonli trapetsiyadan iborat. Trapetsiya asoslari 8 va 18 ga teng.  $ABCD$  va  $A_1B_1C_1D_1$  trapetsiyada  $KN$ ,  $K_1N_1$  o‘rta chiziqlar.



Javoblar.

- A)  $12\sqrt{313}$
- B)  $6\sqrt{313}$
- C)  $265\frac{11}{13}$
- D)  $132\frac{12}{13}$
- E)  $156\sqrt{13}$
- F) 156

33. Prizma diagonal kesimi yuzini toping.

34.  $KK_1N_1N$  kesim yuzini toping.

35.  $LL_1M_1M$  kesim yuzini toping.

36.  $f(x) = x^3 + ax^2 + bx + 1$  tenglamaning nollari,  $r, s, t$  va  $g(x) = x^3 + mx^2 + nx^2 + p$  funksiyaning nollari  $r^2, s^2, t^2$  hamda  $g(-1) = -5$  bo'lsin.

a)  $n - m - p$  ni toping.

Javob: \_\_\_\_\_

b)  $(a - 1)^2 + (b - 1)^2$  ni toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\sin^4 x + 2 \sin^2 2x = 9 \cos^4 x .$$

a)  $\left[-\frac{\pi}{2}; \frac{5\pi}{2}\right]$  oraliqdagi yechimlar sonini toping.

Javob: \_\_\_\_\_

b) Tenglamaning  $[-\pi; \pi]$  oralig'idagi yechimlari yig'indisini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

38.  $f(x)$  kvadrat funksiya uchun  $f(0) = 2$  tenglik o'rinli.

a)  $f(x + 5) = f(x) + x + 1$  bo'lsa,  $f(2025)$  ni toping.

Javob: \_\_\_\_\_

b)  $f(x - 9) = f(x) + x - 5$  bo'lsa,  $f(2025)$  ni toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

39.  $f(x) = x^2 - 6x + 12$  funksiya berilgan.

a)  $f(x)$  funksiyaga o'tkazilgan urinma  $y = 2x - 1$  to'g'ri chiziq bilan umumiy nuqtaga ega bo'lmasa, urinish nuqtasining ordinatasini toping.

Javob: \_\_\_\_\_

b)  $f(x)$  funksiyadagi  $A$  nuqtadan  $y = -2x + 1$  to'g'ri chiziqqa bo'lgan eng yaqin masofani toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

40.  $f(x) = x^2$  va  $f(x) = \sqrt{x}$  funksiyalar berilgan.

a) Bu funksiyalar bilan chegaralangan soha yuzini toping.

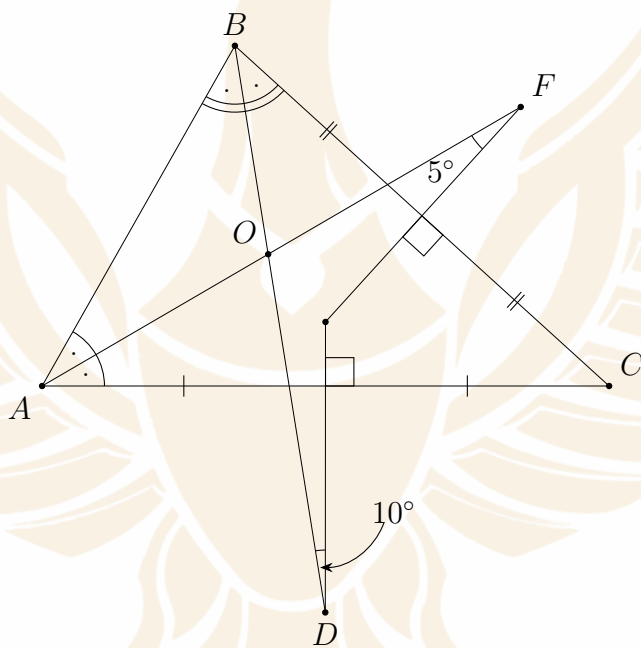
Javob: \_\_\_\_\_

b) Bu funksiyalar bilan chegaralangan soha yuzini  $Ox$  o'qi atrofida  $360^\circ$  aylantirishdan hosil bo'lgan jism hajmini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41.  $ABC$  uchburchakning  $AF$  va  $BD$  bissektissalari o'tkazilgan. Bissektrisalar o'rta perpendikulyarlar bilan  $5^\circ$  va  $10^\circ$  ga teng burchaklar hosil qiladi.  $AC = 3$  bo'lsin.



a)  $\angle ACB$  ni toping.

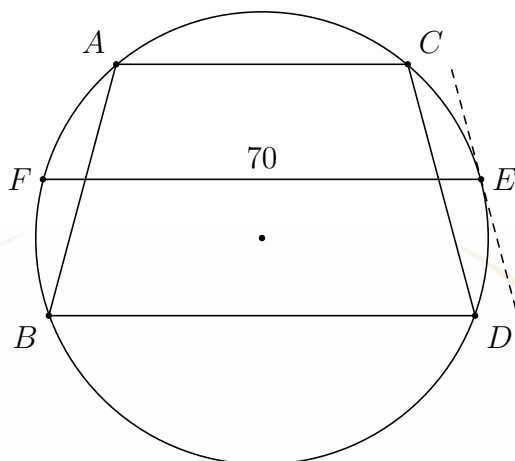
Javob: \_\_\_\_\_

b)  $OF$  ning uzunligini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

42. Aylanaga ichki trapetsiya chizilgan. Trapetsiyaning kichik asosi  $AC = 56$  ga teng. Aylanadan  $E$  va  $F$  nuqtalar olinib trapetsiyaning yon tomoniga parallel to'g'ri chiziqlar o'tkazilgan. Agar  $EF = 70$  va aylananing radiusi  $R = 14\sqrt{7}$  bo'lsa,



a) Trapetsiyaning  $BC$  diagonali uzunligini toping.

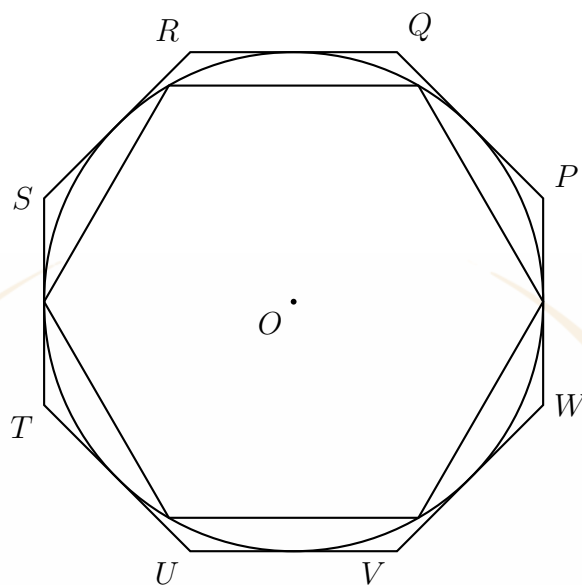
Javob: \_\_\_\_\_

b) Trapetsiya yuzini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

43. Muntazam sakkizburchakka aylana ichki chizilgan va shu aylanaga muntazam oltiburchak ichki chizilgan.



a) Aylana markazida oltiburchak tomonigacha bo'lgan eng qisqa masofaning, aylana markazidan muntazam sakkizburchak tomonigacha bo'lgan eng qisqa masofaga nisbatini toping.

Javob: \_\_\_\_\_

b) Muntazam oltiburchak yuzasining muntazam sakkizburchak yuzasiga nisbatini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

44. Muntazam to'rtburchakli kesik piramidaga shar ichki chizilgan. Piramidaning katta asosi yuzi kichik asosi yuzidan 9 marta katta va uning to'la sirti 312 ga teng bo'lsa,

a) Piramidaning hajmini toping.

Javob: \_\_\_\_\_

b) Shar hajmini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

45. Muntazam to'rtburchakli piramidaning qirrasi uzunligi 2 ga teng. Piramidaning hajmi eng katta bo'lsa,

a) Piramida asosi uzunligini toping.

Javob: \_\_\_\_\_

b) Piramida hajmining eng katta qiymatini toping.

Javob: \_\_\_\_\_

**Diqqat!** Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.